

MASOUD ABDOLLAHI, Ph.D.

[masoudabd.github.io](https://github.com/masoudabd) | masudabdollahi@gmail.com | [Google Scholar](#) | [LinkedIn](#)

Research Scientist with a Ph.D. and 4+ years of industry experience building and deploying machine learning systems at scale. Expert in deep learning (CNNs, RNNs/LSTMs, Transformers), foundation models, vision-language models, and multimodal AI; applied to real manufacturing environments with measurable product impact. Author of 27+ peer-reviewed publications (650+ citations), 5 patent filings (1 published), and a track record of leading research agendas end-to-end from prototype to production. Experienced contributor to the broader research community through publications, peer review, and cross-institutional collaboration.

RESEARCH EXPERTISE

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| • Machine & Deep Learning (CNN, RNN/LSTM, Transformer) | • Foundation Models & Vision-Language Models | • Multimodal & Self-Supervised Learning |
| • Computer Vision & Pose Estimation | • Time Series Classification & Sensor Fusion | • Generative AI & LLM-powered Systems |
| • Human Motion Analysis & NLP | • Data Mining & Experimental Design | • Python, PyTorch, TensorFlow, Azure OpenAI, HuggingFace |

PROFESSIONAL EXPERIENCE

Senior R&D Scientist — Hitachi America, Ltd.

Feb. 2023 – Present

- Promoted to Senior R&D Scientist (Jun 2026); lead multiple R&D projects and own a research roadmap.
- Led end-to-end research and deployment of a Vision-Language Model (VLM) + generative AI system that cut a critical manufacturing bottleneck by **25%** at a U.S. Hitachi facility — from research concept to full production deployment.
- Designed and shipped deep learning models (CNN, RNN-LSTM, Transformer) and classical ML pipelines on real automotive OEM datasets to solve manufacturing quality and warranty prediction challenges.
- Built multimodal perception systems combining wearable sensors and computer vision (pose estimation, action recognition), achieving **>90% accuracy** across real production environments.
- Accumulated **500+ hours/year** of on-site factory data collection — bridging lab research and industrial-scale deployment.
- Inventor on **5 patent filings in AI/ML, generative AI, and sensing for industrial applications** (1 published — US 2026/0198641 A1; 1 under USPTO review; 3 in preparation).
- Collaborated cross-functionally with engineering, IP/legal, and leadership teams to define research roadmap and technical strategy.

Research Engineer III (Part-time) — Oregon Health & Science University

Feb. 2024 – Present

- Applying state-of-the-art wearable and computer vision methods to objectively assess worker performance and safety; developing sensor-based deep learning models for neurological impairment detection.

Graduate Research Assistant — Rochester Institute of Technology (Biomechanics & Ergonomics Lab)

Aug. 2017 – Dec. 2023

- Designed and led a novel data-driven fall risk assessment model for stroke survivors using 8 IMU sensors and clinical cognitive dual-tasking — resulting in multiple top-journal publications and a defended Ph.D. dissertation.
- Developed camera-based fall risk assessment, smartphone fatigue detection (SVM; 91% accuracy), and LBP patient classification (SVM/MLP; 75%) — three independent research agendas led from hypothesis to publication.
- Contributed to NIH proposal writing (R01, R03, R21) and external grants, building research strategy alongside faculty and clinical collaborators.
- Led clinical data collection at Unity Hospital with patients with neurological conditions; mentored junior students and teaching assistants across 10+ semesters.

R&D Scientist Intern — Aster Pet Sciences

Jun. – Aug. 2021

- Led small team to design, collect data, and ship a real-time IMU-based dog activity recognition system (SVM; >70% accuracy). First full ownership of an AI research-to-product pipeline.

SELECTED PUBLICATIONS

27+ peer-reviewed publications (650+ citations) spanning machine learning, deep learning, computer vision, and human motion. Full list: [Google Scholar](#)

- Abdollahi et al. "Smart wearable insoles in industrial environments: A systematic review." *Applied Ergonomics*, 2024. [PDF](#)
- Abdollahi et al. "Fall risk assessment in stroke survivors: a machine learning model using detailed motion data from common clinical tests and motor-cognitive dual-tasking." *Sensors* 24(3), 2024. [PDF](#)
- Jahangiri, Abdollahi et al. "An inpatient fall risk assessment tool: Application of machine learning models on intrinsic and extrinsic risk factors." *Machine Learning with Applications*, 2024. [PDF](#)
- Jahangiri, Abdollahi et al. "A machine learning model to predict heart failure readmission: toward optimal feature set." *Frontiers in Artificial Intelligence*, 2024. [PDF](#)
- Abdollahi et al. "A systematic review of fall risk factors in stroke survivors." *Frontiers in Bioengineering and Biotechnology*, 2022. [PDF](#)
- Abdollahi et al. "Kinematic Analysis of 360° Turning in Stroke Survivors Using Wearable Motion Sensors." *Sensors* 22(1), 2022. [PDF](#)
- Kuber, Abdollahi et al. "A Systematic Review on Evaluation Strategies for Field Assessment of Upper-Body Industrial Exoskeletons." *Annals of Biomedical Engineering*, 2022. [PDF](#)
- Abdollahi et al. "Using a Motion Sensor to Categorize Nonspecific Low Back Pain Patients: A Machine Learning Approach." *Sensors* 20(12), 2020. [PDF](#)
- Karvekar*, Abdollahi* et al. "Smartphone-based human fatigue level detection using machine learning approaches." *Ergonomics*, 2021. (*equal contribution) [PDF](#)

Active peer reviewer: Medical Engineering & Physics, Human-centric Computing & Info Sciences, IEEE Open Journal of Computer Society, Frontiers in Bioengineering & Biotechnology, Ergonomics, Pain Medicine, The Journal of Supercomputing (2017–present).

PATENTS

- Zhou, Q., & Abdollahi, M. Smart Sole Architecture for Measuring Ground Reaction Force and Method for Unifying Coordinates for Feet Positions and Gestures. U.S. Patent Application Publication US 2026/0198641 A1. Published Jul. 16, 2026.
- Abdollahi M. et al. Agentic AI & generative AI system for industrial operations. U.S. Patent Application. Under USPTO Review. Hitachi America, Ltd. (2025).
- Abdollahi M. et al. (3 additional applications in preparation — agentic AI, generative AI, sensor-based human performance assessment & training). Hitachi America, Ltd. IP & Legal Team (2026).

EDUCATION

Ph.D. in Mechanical & Industrial Engineering — Rochester Institute of Technology Aug. 2017 – Dec. 2023
Dissertation: A Comprehensive Machine Learning Approach to Assess Fall Risk in Stroke Survivors: Integrating Common Clinical Tests, Motor-Cognitive Dual-Tasks, Wearable Motion Sensors, and Detailed Motion Analysis

M.Sc. in Mechanical Engineering — Sharif University of Technology Sep. 2012 – Jun. 2015
Ranked 1st among M.Sc. students in Biomechanics department.

B.Sc. in Mechanical Engineering — Semnan University Sep. 2007 – Sep. 2011

HONORS & RECOGNITION

- Ranked 1st in Machine Learning Competition, Dept. of Computer Engineering, RIT (among ~60 students, mostly CS/CE majors) — 2018
- 1 of 10 global travel grants awarded for the Autumn School on Movement Science, Berlin, Germany — 2016
- Ranked 1st among M.Sc. students in Biomechanics, Sharif University of Technology — 2015